

ChemVault Dependencies & Setup Report

A comprehensive catalogue of all frontend, backend, Python packages, and binary tools

1. Backend NodeJS Production Packages

Package Name	Installation/Run Command	Purpose & Role in ChemVault
@sendgrid/mail	<code>npm install @sendgrid/mail</code>	Sends email notifications via SendGrid service API.
axios	<code>npm install axios</code>	Performs HTTP requests to communicate with external APIs (UniProt, PDB, etc.).
axios-retry	<code>npm install axios-retry</code>	Automatically retries failed external API requests using exponential backoff.
bcryptjs	<code>npm install bcryptjs</code>	Secures user account passwords using salt and blowfish encryption hashing.
cors	<code>npm install cors</code>	Configures CORS rules to accept secure requests from the React frontend.
dotenv	<code>npm install dotenv</code>	Loads configuration credentials and environment variables from a local .env file.
express	<code>npm install express</code>	Core backend web framework for defining routes, endpoints, and middleware.
express-session	<code>npm install express-session</code>	Handles persistent session records for user login states.
form-data	<code>npm install form-data</code>	Constructs and dispatches multi-part form requests (e.g. for docking submissions).
jsonwebtoken	<code>npm install jsonwebtoken</code>	Signs, encodes, and verifies JWT tokens to secure REST API endpoints.
lowdb	<code>npm install lowdb@1.0.0</code>	Local flat JSON file database used to store users and workspace records.
multer	<code>npm install multer</code>	Processes incoming multipart/form-data for files uploaded to the server.
nodemailer	<code>npm install nodemailer</code>	Connects to SMTP servers to deliver emails from contact form submissions.
twilio	<code>npm install twilio</code>	Orchestrates SMS transmission for mobile text-based alerts.
xlsx	<code>npm install xlsx</code>	Parses and outputs spreadsheet rows from uploaded Excel sheets.

2. Frontend NodeJS Production Packages

Package Name	Installation/Run Command	Purpose & Role in ChemVault
react	<code>npm install react</code>	Foundation UI component building library.

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react-dom	<code>npm install react-dom</code>	Renders and binds React nodes onto browser webpage targets.
react-router-dom	<code>npm install react-router-dom</code>	Manages page navigation, pathways, and state history.
axios	<code>npm install axios</code>	Enables request dispatch to the NodeJS Express backend service.
lucide-react	<code>npm install lucide-react</code>	Supplies custom lightweight SVG vector design icons for the interface.

3. Frontend NodeJS Development Packages

Package Name	Installation/Run Command	Purpose & Role in ChemVault
vite	<code>npm install --save-dev vite</code>	High-performance build tool, module bundler, and dev server.
@vitejs/plugin-react	<code>npm install --save-dev @vitejs/plugin-react</code>	Vite extension enabling hot-reloading for React modules.
tailwindcss	<code>npm install --save-dev tailwindcss</code>	Utility-first CSS framework for responsive UI design.
postcss	<code>npm install --save-dev postcss</code>	CSS post-processing engine for Tailwind directives.
autoprefixer	<code>npm install --save-dev autoprefixer</code>	Injects browser-specific CSS vendor prefixes automatically.
eslint	<code>npm install --save-dev eslint</code>	Statically analyzes React files for code consistency and error detection.

4. Frontend CDN-Loaded Web Packages

Package Name	Installation/Run Command	Purpose & Role in ChemVault
jspdf		Direct client-side PDF document generation from JSON/text.
jspdf-autotable		Draws tabular datasets inside generated PDF reports.
xlsx		Enables browser-based Excel file creation and exports.
jszip		Zips workspace logs and coordinates for convenient single-file downloads.

5. Python Backend Scientific Packages

Package Name	Installation/Run Command	Purpose & Role in ChemVault
biopython	<code>pip install biopython</code>	Parses and manipulates protein coordinate data from standard PDB files.
pdbfixer	<code>pip install pdbfixer</code>	Repairs broken protein models by inserting missing atoms, loops, and residues.

Package Name	Installation/Run Command	Purpose & Role in ChemVault
openmm	<code>pip install openmm</code>	High-performance molecular simulation engine used to manage atomic coordinates.
numpy	<code>pip install numpy</code>	Processes dimensional vector coordinates, grids, and bounding boxes.
pandas	<code>pip install pandas</code>	Manages tabular summaries of docking energy scores and interactions.
matplotlib	<code>pip install matplotlib</code>	Plots static binding configurations, grids, and docking charts.
seaborn	<code>pip install seaborn</code>	Generates clean, styled statistical data visualizations.

6. Command-Line Binaries & Engines

Package Name	Installation/Run Command	Purpose & Role in ChemVault
AutoDock Vina	Run: <code>./vina.exe --help</code>	Molecular docking engine doing binding score calculations.
Open Babel	Run: <code>obabel --help</code>	Cleans structures, generates 3D geometries, and converts formats (.sdf to .pdbqt).
Fpocket	Run: <code>fpocket -f protein.pdb</code>	Detects binding pockets and potential active sites on protein structures.